

FUNCTIONS AND GRAPHS, (SKETCHING QUADRATIC, CUBIC AND RECIPROCAL

Name: _____

Date: _____

LO: I can sketch quadratic, cubic and reciprocal graphs from their equations.

Types of graph

A **quadratic** graph has the form $y = ax^2 + bx + c$ and is a **U-shape (parabola)**. A **cubic** has the form $y = ax^3$ and has an S-shaped curve.

A **reciprocal** graph has the form $y = \frac{a}{x}$ and has two separate curves (hyperbola) that never touch the axes.

Recognising graphs

●	$y = x^2 - 3$	quadratic, U-shape shifted down 3
●	$y = x^3$	cubic, S-shape through origin
●	$y = 1/x$	reciprocal, two curves in opposite quadrants
●	$y = x^2$ is a straight line	it is a parabola

Key idea

To sketch a graph, find key points (intercepts, turning points) and know the general shape from the highest power of x .

$$y = \text{ax}^2 + bx + c \quad \text{parabola}$$

The highest power tells you the shape

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - quadratic graph

Sketch $y = x^2 - 4$

Find y-intercept: when $x = 0$, $y = -4$
Find x-intercepts: $x^2 - 4 = 0$, so $x = 2$
or $x = -2$

Minimum point at $(0, -4)$; U-shape opening upwards

U-shaped parabola crossing x-axis at $(-2, 0)$ and $(2, 0)$, vertex at $(0, -4)$

Plot the intercepts and vertex, then draw a smooth curve

Example 2 - negative quadratic

Sketch $y = -x^2 + 9$

y-intercept: when $x = 0$, $y = 9$
x-intercepts: $-x^2 + 9 = 0$, $x^2 = 9$, $x = \pm 3$

Maximum at $(0, 9)$; inverted U-shape

Inverted parabola with maximum at $(0, 9)$, crossing x-axis at $(-3, 0)$ and $(3, 0)$

Negative coefficient of x^2 means the parabola opens downwards

Example 3 - cubic graph

Sketch $y = x^3 - x$

Factorise: $y = x(x^2 - 1) = x(x + 1)(x - 1)$
x-intercepts at $x = -1, 0, 1$

S-shaped curve passing through all three roots

S-shaped cubic crossing x-axis at $-1, 0$, and 1

Factorising a cubic helps find the roots for sketching.

Example 4 - reciprocal graph

Sketch $y = 3/x$

When $x = 1$, $y = 3$; when $x = 3$, $y = 1$;
when $x = -1$, $y = -3$

As $x \rightarrow 0$, $y \rightarrow \infty$; as $x \rightarrow \infty$, $y \rightarrow 0$

Two curves: one in quadrant 1 and one in quadrant 3

Hyperbola with curves in quadrants 1 and 3, asymptotes along both axes

The axes are asymptotes; the graph never touches or crosses them

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying graph types

State whether each equation gives a quadratic, cubic or reciprocal graph.

- a) $y = 2x^2 + 1$
- b) $y = x^3 - 2x$
- c) $y = 5/x$
- d) $y = -x^2 + 4x - 3$

2. Finding intercepts

Find the y-intercept and x-intercepts for each quadratic.

- a) $y = x^2 - 9$
- b) $y = x^2 - 5x + 6$
- c) $y = x^2 + 2x - 8$
- d) $y = -x^2 + 4$

3. Cubic graphs

Find the x-intercepts of each cubic by factorising.

- a) $y = x^3 - 4x$
- b) $y = x^3 - 9x$
- c) $y = x^3 + x^2 - 6x$
- d) $y = 2x^3 - 8x$

4. Reciprocal values

For $y = 6/x$, find y for each value of x.

- a) $x = 1$
- b) $x = 2$
- c) $x = 6$
- d) $x = -3$

5. Mixed sketching

Describe the key features of each graph.

- a) $y = x^2 - 1$
- b) $y = -x^3$
- c) $y = 4/x$
- d) $y = (x - 2)^2$
- e) $y = x^3 - x$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $y = x^2$ produces a U-shaped parabola ☐

Correct answer: _____

Reason: _____

b) $y = 1/x$ passes through the origin ☐

Correct answer: _____

Reason: _____

c) A cubic graph can cross the x-axis up to 3 times ☐

Correct answer: _____

Reason: _____

d) $y = -x^2$ opens upwards ☐

Correct answer: _____

Reason: _____

e) The reciprocal graph $y = a/x$ has asymptotes at $x = 0$ and $y = 0$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A ball is thrown upwards. Its height h metres after t seconds is given by $h = 20t - 5t^2$. Sketch the graph. What is the maximum height and when does the ball land?

Expression: _____

Simplified: _____

b) The time T hours to travel 120 miles at speed v mph is given by $T = 120/v$. Sketch the graph for v from 20 to 120. How does doubling the speed affect the journey time?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

